

MODULE 3

- Basic principles
- Elements of feedback loop
- Block diagram
- Control performance measures for common input changes
- Selection of variables for control approach to process control
- Controller modes ————— discrete - ON/OFF, multi-position
continuous - P, PI, PID
- Tuning parameters
- Process step testing
- Tuning for Quarter decay ratio resp.
- Minimal error integral criteria
- Sampled data controllers
- Controller tuning for integrating processes -
↳ model of liquid level control system.

DISCONTINUOUS CONTROLLER MODES

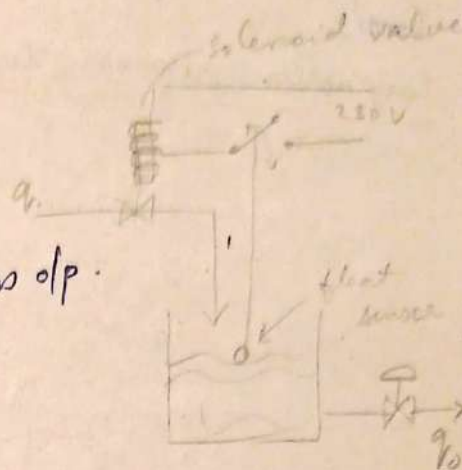
- also called "switching controllers"

ON-OFF CONTROLLERS

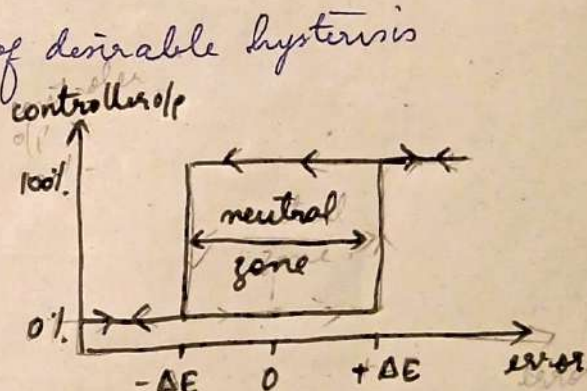
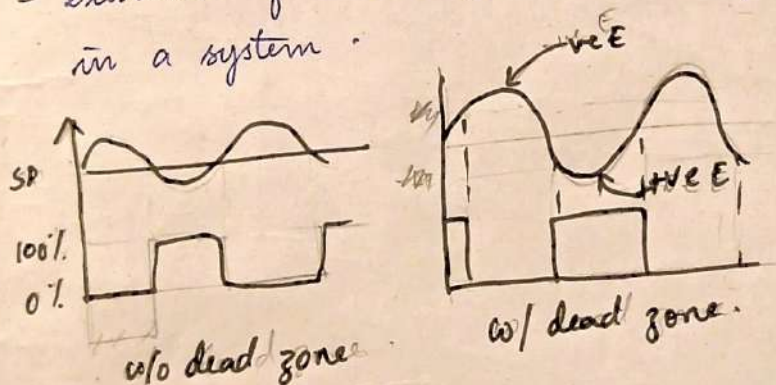
or two-position controller

$$P = \begin{cases} 0 & e < 0 \\ 100\% & e > 0 \end{cases}$$

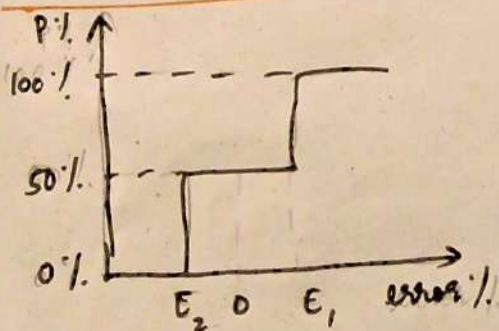
P: process o/p.



- signal $> SP$, controller: 0. & vice versa.
- continues indefinitely.
- oscillations increase in amplitude and frequency if process measurement is more responsive.
- usually operated w/ "neutral zone" or "dead zone" around SP.
- control action taken only if controlled variable ~~goes~~ goes beyond this dead zone.
- existence of neutral zone is example of desirable hysteresis in a system.



MULTI POSITION CONTROLLER



error b/w E_2 & E_1 : process o/p is 50%.

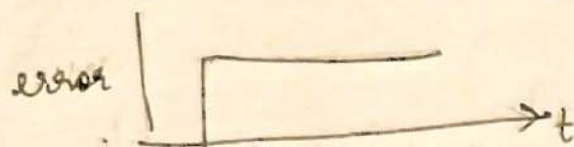
error exceeds E_1 : process o/p is 100%.

can be thought of having 3 states.

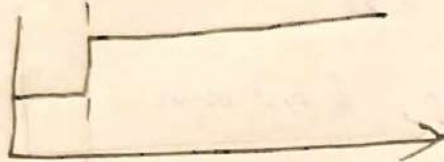
eg: hot case of heating system, 3 states are ON, OFF & cooling.

Good

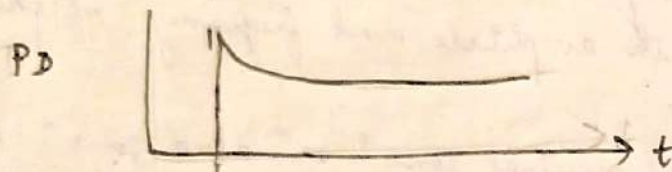
Controller response to step input:



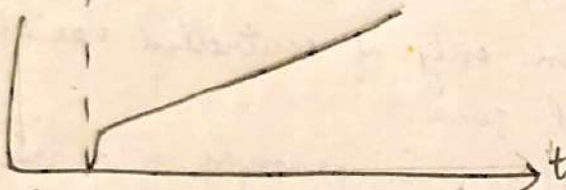
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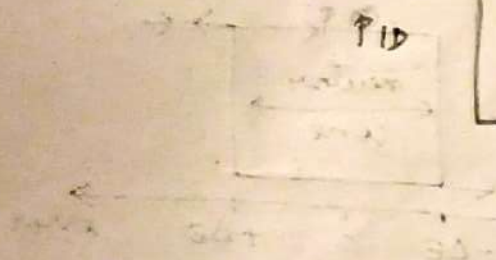
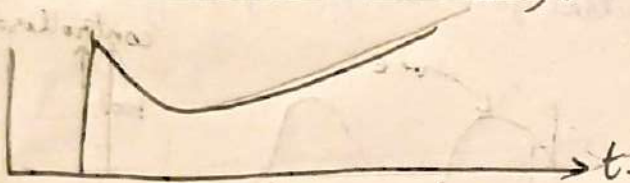
P



P



PID



ZIEGLER-NICHOLS METHOD

- adjusting a closed loop until steady oscillations occur.
- controller settings are based on conditions that generate oscillations
- based on freq. resp. analysis.
- closed loop procedure.

Steps

- 1) Place controller in closed loop of low gain (~~K_p~~).
- 2) Disable integral and derivative action
- 3) using P control only, w/ ffb loop closed, introduce a SP change and vary proportional gain (K_p) until sys. oscillates continuously.
 freq. of conts oscillations is cross over frequency (ω_{co}) and K_u is (ultimate gain) gain which causes sys. to oscillate in marginally stable system.
 ultimate period, $P_u = \frac{2\pi}{\omega_{co}}$

~~Notes~~ Tuning parameters for according to Ziegler-Nichols' method.

	K_p	T_i	T_d
P	$0.5 K_u$		
PI	$0.45 K_u$	$\frac{P_u}{1.2}$	
PID	$0.6 K_u$	$\frac{P_u}{1.2}$	$\frac{P_u}{8}$

- * For PI control, use lower Proportional gain bc presence of integral control mode introduces additional phase lag in all frequencies which destabilizes system.
- * For PID control, since derivative control introduces phase lead w/ strong stabilizing effects in CL resp. so proportional gain can be higher in PID controllers w/o destabilizing system.

- K_p should be increased in very small steps.
- Tuning should be only done after ensuring controller is in operating range.
- If K_p is $\uparrow$ ed in large steps, gain will increase very quickly leading to limit cycles which controller act as ON-OFF controller.
- This also creates sustained oscillations called limit cycles.
- Limit cycles must not be confused with sustained osc. we need.
- We need $A_p = 1$ but $A_p \neq 1$ for limit cycles.

MODELLING OF LIQUID LEVEL SYSTEMS

Mass balance eqn.

$$\frac{\text{Mass into sys}}{\text{sec}} - \frac{\text{Mass out of sys}}{\text{sec}}$$

= time rate of change of mass accumulation

$$q_i(t) - q_o(t) \propto \frac{d}{dt} h(t)$$

$$q_i(t) - q_o(t) =$$

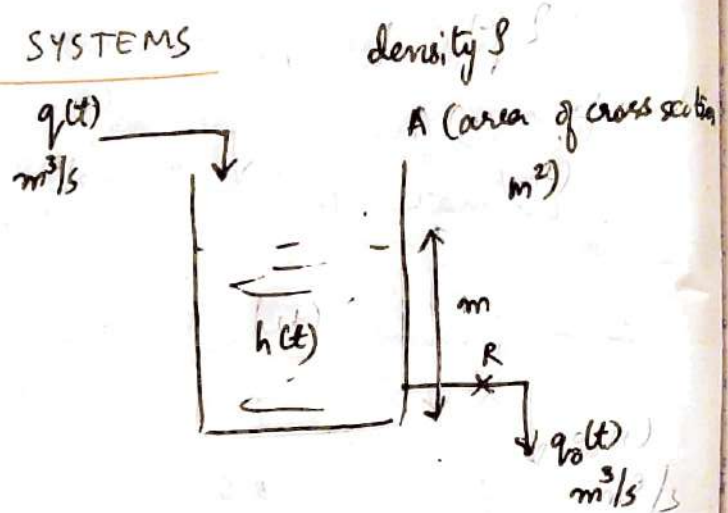
$$\rho q_i(t) - \rho q_o(t) = \rho A \frac{d}{dt} h(t)$$

$$q_i(t) - q_o(t) = A \frac{d}{dt} h(t) \quad \text{--- (1)}$$

$$q_o(t) = \frac{h(t)}{R} \quad \text{--- (2)}$$

at steady state: $\text{from (1)} \Rightarrow q_{is} - q_{os} = 0, \quad q_{os} = \frac{h_s}{R} \quad \text{--- (4)}$

bc $h(t)$ doesn't change



we want relations b/w $h(t)$ & $q(t)$

$$\textcircled{1} - \textcircled{3} \Rightarrow q_i(t) - q_{i,s} - q_o(t) + q_{o,s} = A \frac{d}{dt} h(t) \quad \text{deviation variables}$$

$$Q_i(t) - Q_o(t) = A \frac{d}{dt} H(t) \quad \text{--- } \textcircled{5}$$

{changing $q_i(t)$ to $q(t)$ }

$$\textcircled{2} - \textcircled{4} \Rightarrow Q_o(t) = \frac{H(t)}{R} \quad \text{--- } \textcircled{6}$$

$$\text{LT of } \textcircled{5} \text{ \& } \textcircled{6} \Rightarrow Q_o(s) - Q_o(s) = A_s H(s) ; \quad Q_o(s) = \frac{H(s)}{R}$$

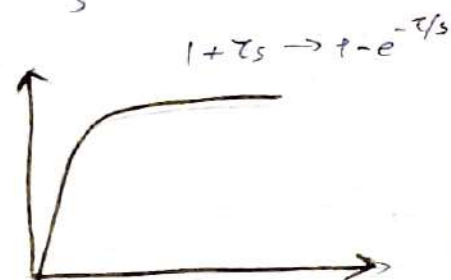
$$\Rightarrow Q(s) - \frac{H(s)}{R} = A_s H(s) \Rightarrow H(s) \left(A_s + \frac{1}{R} \right) = Q(s)$$

$$\boxed{\frac{H(s)}{Q(s)} = \frac{R}{1 + RA_s}} \quad \text{of form } \frac{K}{\tau s + 1} ; \quad \begin{cases} \tau = RA \\ K = R \end{cases}$$

to observe change, apply ~~sp~~ step input, i.e. $Q(s) = \frac{A}{s}$.

$$\Rightarrow H(s) = \frac{K}{1 + \tau s} \cdot \frac{A}{s} \Rightarrow \frac{KA}{s} - \frac{KA}{1 + \tau s}$$

$$\Rightarrow h(t) = KA(1 - e^{-t/\tau})$$



Since system will come to new ss,
this is a self regulating system.

[as if $t \uparrow$, pressure head $(h(t)) \uparrow$, $\therefore d/t \uparrow$]

Note: this model doesn't account for downstream head (u)

$$\text{if } u \text{ were present, eq } \textcircled{2} \Rightarrow q_o = \frac{h-u}{R}$$

$$\text{and } \Rightarrow H(s) = \frac{R Q(s)}{1 + RA_s} + \frac{U(s)}{1 + RA_s}$$

$$\Rightarrow \frac{R Q(s)}{1 + \tau s} + \frac{U(s)}{1 + \tau s} \quad , \quad \tau = RA$$

Non interacting 1st order systems

q(u)



relation b/w

$H_2(s)$ & $Q(s)$.

$$\frac{H_2}{Q} = \frac{H_2}{Q_1} \cdot \frac{Q_1}{Q_2}$$

Tank 1: $\frac{H_1(s)}{Q(s)} = \frac{R_1}{1 + A_1 R_1 s}$ Tank 2: $\frac{H_2(s)}{Q_1(s)} = \frac{R_2}{1 + A_2 R_2 s}$

Tank 1:

$$Q(t) - Q_1(t) = A_1 \frac{d}{dt} R_1 Q_1(t) \quad (H_1(t) = R_1 Q_1(t))$$

$$\frac{Q_1(s)}{Q(s)} = \frac{1}{1 + A_1 R_1 s}, \quad \tau_1 = A_1 R_1$$

Tank 2:

$$Q_1(t) - Q_0(t) = A_2 \frac{d}{dt} H_2(t) \quad ; \quad Q_0(t) = R_2 \frac{H_2(t)}{R_2}$$

$$\Rightarrow Q_1(t) - \frac{H_2(t)}{R_2} = A_2 \frac{d}{dt} H_2(t)$$

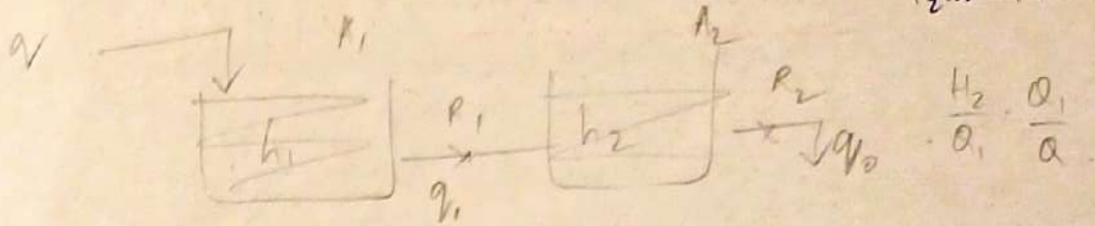
$$\frac{H_2(s)}{Q_1(s)} = \frac{R_2}{1 + A_2 R_2 s}, \quad \tau_2 = A_2 R_2$$

$$\frac{H_2(s)}{Q(s)} = \frac{R_2}{(1 + \tau_1 s)(1 + \tau_2 s)}$$

$$\left[\frac{H_2(s)}{Q(s)} = \frac{R}{1 + (\tau_1 + \tau_2)s + \tau_1 \tau_2} \right]$$

Interacting 1st order system

relation b/w $H_2(s)$ & $Q(s)$



Tank 1: $Q(s) - Q_1(s) = A_1 \frac{d}{dt} H_1(t)$ — ①

$Q_1(s) = \frac{H_1(s) - H_2(s)}{R_1} \Rightarrow H_1(s) - H_2(s) = Q_1(s) R_1$ — ②

$Q(s) - Q_1(s) = A_1 \frac{d}{dt} [R_1 Q_1(s) + H_2(s)]$

$\Rightarrow Q(s) - Q_1(s) = A_1 R_1 Q_1(s) + A_1 H_2(s)$

$\Rightarrow Q(s) - (1 + A_1 R_1 s) Q_1(s) = A_1 H_2(s) ; \tau_1 = A_1 R_1$ — ③

Tank 2: $Q_1(s) - Q_2(s) = A_2 \frac{d}{dt} H_2(t)$

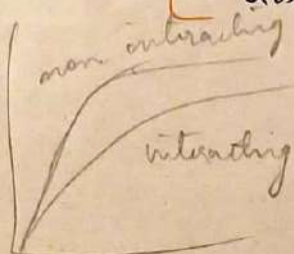
$Q_2(s) = \frac{H_2(s)}{R_2} \therefore Q_1(s) - \frac{H_2(s)}{R_2} = A_2 s H_2(s)$

$\Rightarrow Q_1(s) = H_2(s) \left(\frac{R_2 A_2 s + 1}{R_2} \right)$ — ④

in ③ $\Rightarrow Q(s) - (1 + \tau_1 s) H_2(s) \left(\frac{\tau_2 s + 1}{R_2} \right) = A_1 s H_2(s)$

$\Rightarrow Q(s) = \left(\frac{(1 + \tau_1 s)(1 + \tau_2 s) + A_1 R_2 s}{R_2} \right) H_2(s)$

$\left[\frac{H_2(s)}{Q(s)} = \frac{R_2}{A_1 R_2 s + (1 + \tau_1 s)(1 + \tau_2 s)} \right]$



denominator: $s^2 \tau_1 \tau_2 + s(\tau_1 + \tau_2 + A_1 R_2) + 1$

$A_1 R_2$ term in denominator makes system sluggish

CONTROL PERFORMANCE MEASURES FOR COMMON INPUT CHANGES

just write down name/heading & explanation w/ pencil
& eqn/diagram w/ pen next to it
explanⁿ should be insignificant / do the side

- Set point input changes
- disturbance input changes.

Set point input changes.

changes to set point

- offset: diff. b/w final ss. value of SP and controlled variable.
- Rise time: time from step change till conⁿ reaches new SP.

- Integral error measures.

IAE: absolute value

$$= \int_0^{\infty} |SP - CV| dt$$

ISE: squared error

$$= \int_0^{\infty} [SP - CV]^2 dt$$

ITAE: time product of abs.

$$\text{time \& abs. error} = \int_0^{\infty} t |SP - CV| dt$$

easy to
visually
analyse

when ^{large} deviation
cause more degradation
than smaller ones

penalises errors that persist
for longer

- Decay ratio: ratios of neighbouring peaks.
- Period of oscillation of transient response.
- Settling time: time taken to settle b/w $\pm 5\%$ of desired value.

- Disturbance input changes.

Step disturbance

- infrequent & sudden
- due to changes in other parts of plant

Max deviation

- max deviatⁿ of CV from SP.

Stochastic inputs

- random, continuous stream of small & large disturbances
- process w/ such i/p's called stochastic process.
- major control performance: var: $\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (\bar{CV} - CV_i)^2$

$$\bar{CV} = \frac{1}{n} \sum_{i=1}^n CV_i$$

Sine inputs

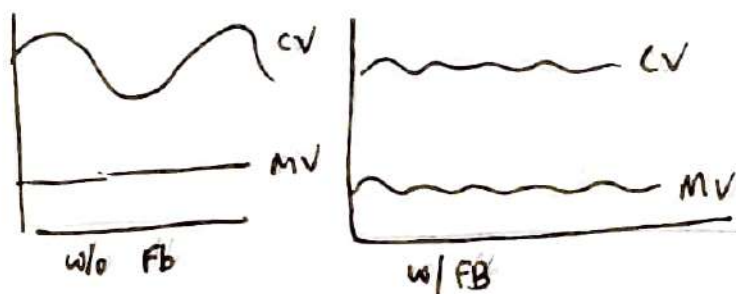
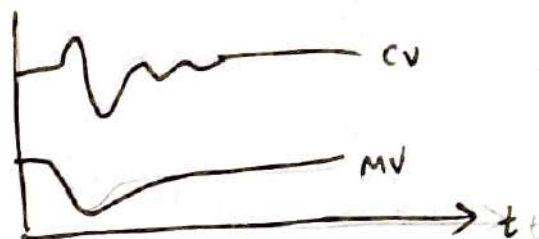
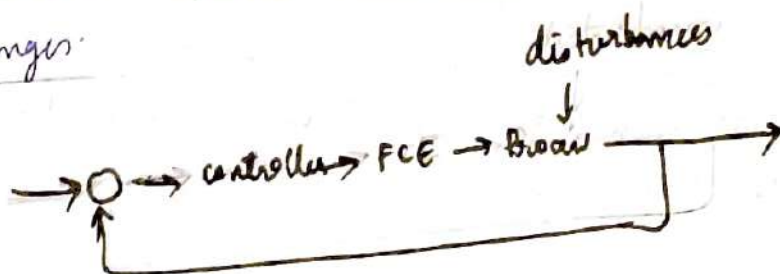
- type of stochastic input

* Feedback reduces variability of controlled variable at expense of $\uparrow$ ed variability of MV.

SELECTION OF VARIABLES FOR CONTROL

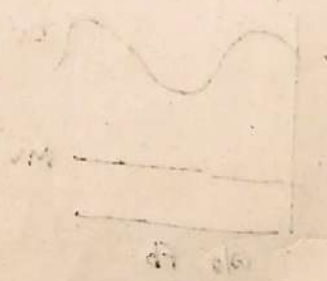
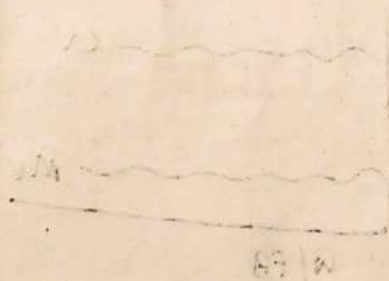
selected based on control objectives:

safety, env'tal protectⁿ, equipment protectⁿ, product quality, profit optimisation, monitoring & diagnosis.



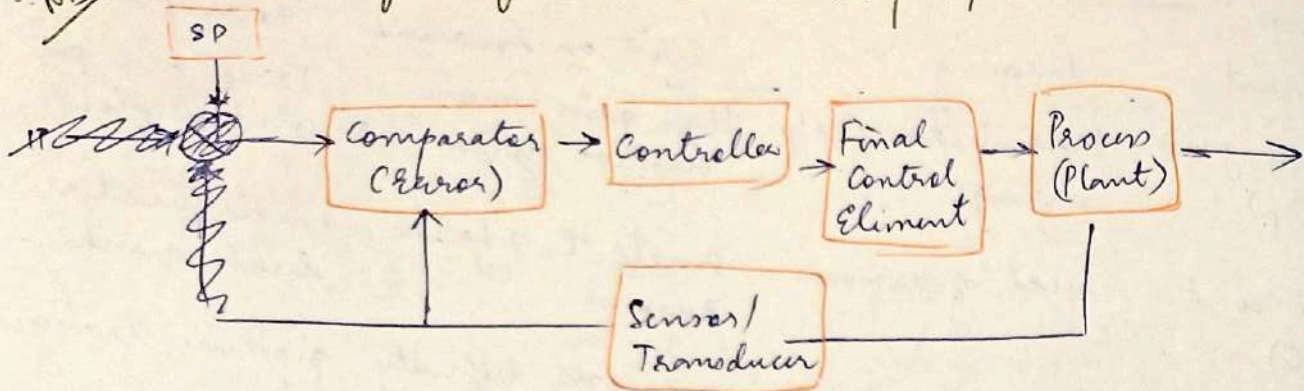
* criteria

- causal relationship b/w ^{→ or valve} value & controlled variable
- automated valve to influence the selected flow.
- fast speed of resp.
- ability to compensate for large disturbances.
- ability to adjust manipulated variable rapidly & w/ little upset to remainder of plant.
- only adjustable variables (flow, conc., level, temp, etc) are
- can be chosen as M.V.



15a) Elements of a feedback control loop ~~at~~

(7)



- SP: Desired value of PV.
- Comparator: Calculates error by comparing measured value w/ SP.
 → determines corrective action needed to minimize this difference
- Controller: performs corrective action
- Final control element: implements controller's corrective action decided by controller
- Process (Plant): physical system being controlled
 - transforms raw materials into products / outputs
- Measurement system
 - instruments / sensors / transducers that monitor PV like temp, pressure, etc flow / level, etc.
 - converts physical quantities into electrical / pneumatic signals for analysis

MOD 4 MOD 5
156) Controller tuning. Need. Open loop tuning of PID controller (7) M3

- adjusting controller parameters so system response is as needed.

performance criteria:

↓ error, short settling time, etc.

Steady State performance criteria

- zero steady state error.

Dynamic performance criteria

↳ simpler using few points

↳ complex using entire closed loop response from $t=0$ till response settles down.

~~eg: overshoot, rise time~~

eg: overshoot, rise time, settling time, decay ratio,

freq. of oscillation of transient response

- common criteria: adjust parameters so that controller system response curve has a decay ratio of $1/4$

i.e. ratio of 1st overshoot of 1st peak to 2nd peak = 4:1

↳ good compromise b/w rapid initial response & quick settling time.

Some things for to get well-tuned controllers:

* Fast resp. to SP changes

* minimal overshoot

* fast settling times

* Fast recovery from disturbances (random)

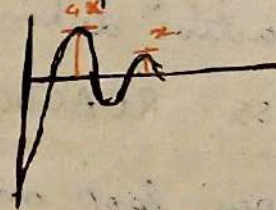
↳ integral squared error (ISE)

settling time

rise time

overshoot

freq. of oscillations of transient



Need/Objective: make sys. stable
 minimize steady state error
 reduce oscillations
 achieve fast & smooth response
 properly handle disturbances

Eqn for PID algorithm.

$$u(t) = K_c \left[e(t) + \frac{1}{T_i} \int e(t') dt' + T_d \left(\frac{de(t)}{dt} \right) \right] + b$$

u : controller signal.

e : error ($|PV - SP|$)

K_c : ~~pro~~ P controller gain

T_i = integral controller parameter

T_d : derivative controller parameter

t : time taken for measurement

b : SP of s/g also called bias / offset

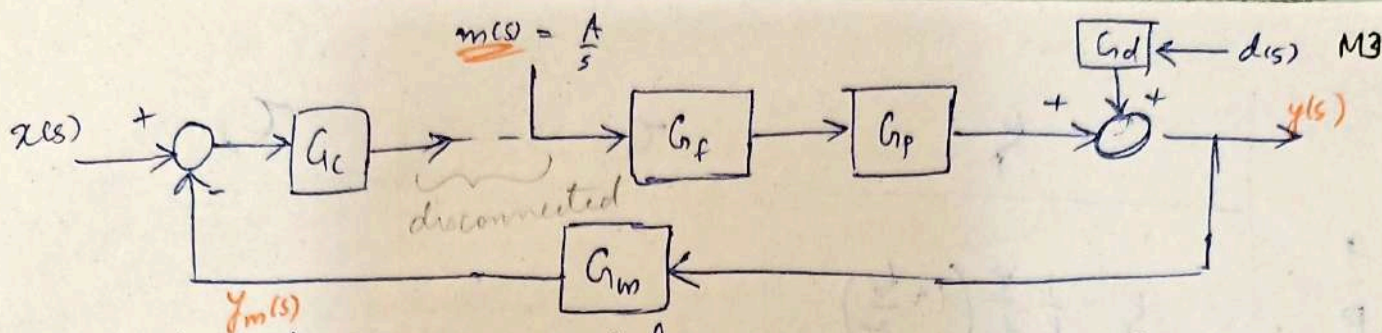
Open loop tuning of PID controller

↳ analyses process behaviour w/o feedback control during active during the test

PROCESS REACTION CURVE / COHEN-COON METHOD

- open loop method

- 1) Disconnect process/plant from feedback control loop.
 - 2) Bring system to middle of operating range.
 - 3) Apply step change at input & observe/plot o/p of process w.r.t time. The curve thus obtained is called the process reaction curve (PRC)
 - 4) PRC gives plant behaviour. ~~T_d, T, K~~
- t_d, T & K can be obtained from PRC.



The feedback loop is disconnected,
Now, controller is not connected to FCE (final control element).

* $m(s)$ actuates FCE.
give step input ($A(s)$ mag: A) to G_f .
This value of op is recorded over time.

$y_m(t)$ is PRC

$$G_{PRC}(s) = \frac{y_m(s)}{m(s)}$$

$$G_{PRC}(s) = \frac{y_m(s)}{m(s)} = G_f G_p G_m \leftarrow \text{all in } s \text{ domain}$$

observed that response to input ~~due~~ step input change has sigmoid shape. which can be approximated by FOPDT.
FOPDT: First Order system (process?) with Dead Time.

$$G_{PRC}(s) = \frac{K e^{-t_d s}}{1 + \tau_s s}$$

$$K = \frac{\text{output at steady state}}{\text{input at steady state}} = \frac{B}{A}$$

$$\tau = \frac{B}{s}, \quad s: \text{ slope of sigmoid response at point of inflection}$$

t_d = time elapsed till system responds.

Steps

- 1) Let procn reach steady state
- 2) switch controller to manual mode
- 3) introduce small step change in controller op
- 4) Plot transient response i.e. of wt time
- 5) ~~curve~~ - curve obtained is PRC.
- 5) Tangent at point of max. slope gives
 t_d : time elapsed b/w till system responds
 τ : 1st order time constant
 K : steady state procn gain
- 6) Controller settings are found from std table.

	K_p	τ_i	τ_d
P	$K_p = \frac{1}{K} \frac{\tau}{t_d} \left(1 + \frac{t_d}{3\tau}\right)$		
PI	$K_p = \frac{1}{K} \frac{\tau}{t_d} \left[0.9 + \frac{t_d}{12\tau}\right]$	$\tau_i = t_d \left[\frac{30 + 3 \frac{t_d}{\tau}}{9 + 20 \frac{t_d}{\tau}} \right]$	
PID	$K_p = \frac{1}{K} \frac{\tau}{t_d} \left[\frac{3}{4} + \frac{t_d}{4\tau} \right]$	$\tau_i = t_d \left[\frac{32 + 6 \frac{t_d}{\tau}}{13 + 8 \frac{t_d}{\tau}} \right]$	$\tau_d = t_d \left[\frac{4}{11 + 2 \frac{t_d}{\tau}} \right]$

gain of PI - con. < P contr.

↳ $\therefore$ integral action makes system more sensitive
(can lead to instability) $\therefore$ ~~gain needs~~

↳ $\therefore$ gain needs to be less.

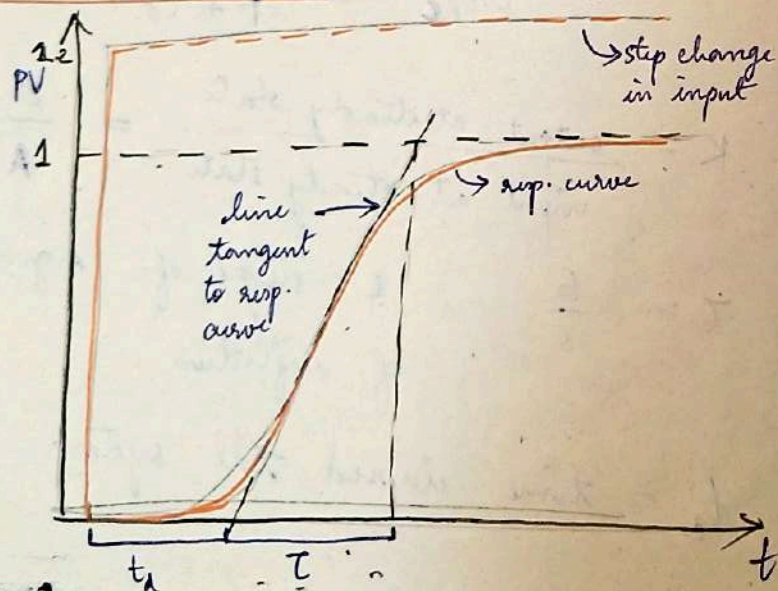
gain of PID contr. > P & PI contr.

↳ $\therefore$ stabilizing effect of derivative control mode.

If $t_d < \tau$, FOPDT may be treated as FOP.

OPTIMUM CONTROLLER SETTING FOR VARIOUS PERFORMANCE.

- 1) Quarter decay ratio
- 2) minimum offset
- 3) Minimum ISE



16 a) P, PI, PID controllers w/ mathematical eqn & eq. of input-output response diagram. CONTS MODE CONTROLLERS. (12) MB

P controller

~~$$m(t) = K_p e(t)$$~~

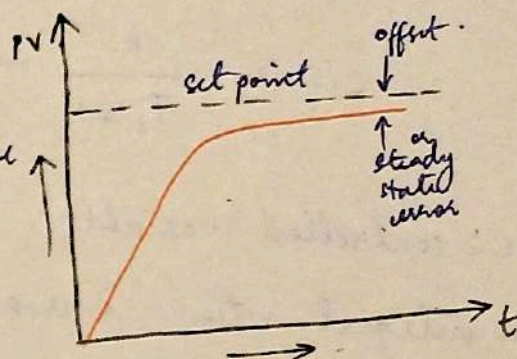
$$U(t) = K_p e(t) + U_0$$

K_p : proportional gain

$e(t)$: error at any instance of time

$U_0 \equiv \text{Bias}$

$U(t)$: controller output



- OFF/ON control oscillates abt SP but PI controller

- P control will always have a steady state error, which can't be removed.

- By $\uparrow K_p$, we can $\downarrow U_0$.

- But beyond a point K_p can't be $\uparrow$, $\therefore$ it'll act like ON/OFF control

- i.e. it'll start oscillating abt SP.

$\Rightarrow$ op will have small +ve $e(t)$ then small -ve $e(t)$, etc.

$$\text{Proportional band PB\%} = \frac{100}{K_p} \%$$

PB% is error needed to change $U(t)$ from min to max (0% to 100%)

of SP.

wide PB, $\Rightarrow \downarrow K_p \Rightarrow$ large error needed $\Rightarrow$ loose control

narrow PB $\Rightarrow \uparrow K_p \Rightarrow$ small error variations $\Rightarrow$ tight control.

PI - controller

$$u(t) = K_p e(t) + \frac{K_p}{T_i} \int e(t) dt + u_0.$$

T_i : integral time

$$C(s) = \frac{R}{T_s + 1} m(s) + \frac{1}{T_s + 1} u(s).$$

C : controlled variable.

- integral action increases order of system (1 to 2)
- integral ~~force~~ action forces controlled variable to return to SP whenever there is a change in load,
- eliminates offset completely.
- but it could also lead to oscillations (of sys.) around set point.
- stabilization time $\uparrow$ due to $\uparrow$ ed order of system.
- integral term is equivalent to adding all errors from $t=0$
 - $\hookrightarrow$ accumulation of all errors.
- after a point, integral term grows & it contributes to output.
- If o/p $>$ SP, $e < 0$, so $\int$ term $\downarrow$ so o/p $\downarrow$
- the integral term is a pole at origin & pole at origin $\downarrow \downarrow$ e_{ss} .

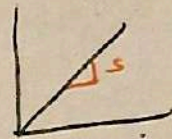
PID - Controller

M3

$$U(t) = K_p e(t) + \frac{K_p}{T_i} \int e(t) dt + K_p T_d \frac{de(t)}{dt} + U_0$$

T_d : derivative time

- derivative term calculates slope of $e(t)$ i.e. $\frac{de(t)}{dt}$



- as slope $\uparrow$ ($\Rightarrow e(t) \uparrow$ quickly),

- $\frac{d}{dt}$ term's contribution $\uparrow$ so it'll take a corrective action accordingly.

Digital PID algorithm

position algorithm:

$$U_n = K_p e_n + \frac{K_p}{T_i} \sum_{i=1}^n e_{i-1} \Delta t + K_p T_d \frac{(e_n - e_{n-1})}{\Delta t} + U_0$$

velocity algorithm:

$$\Delta U_n = K_p \Delta e_n + \frac{K_p}{T_i} e_n \Delta t + K_p T_d \frac{(e_n - 2e_{n-1} + e_{n-2}))}{\Delta t} + U_0$$

* position algo. has reset windup.

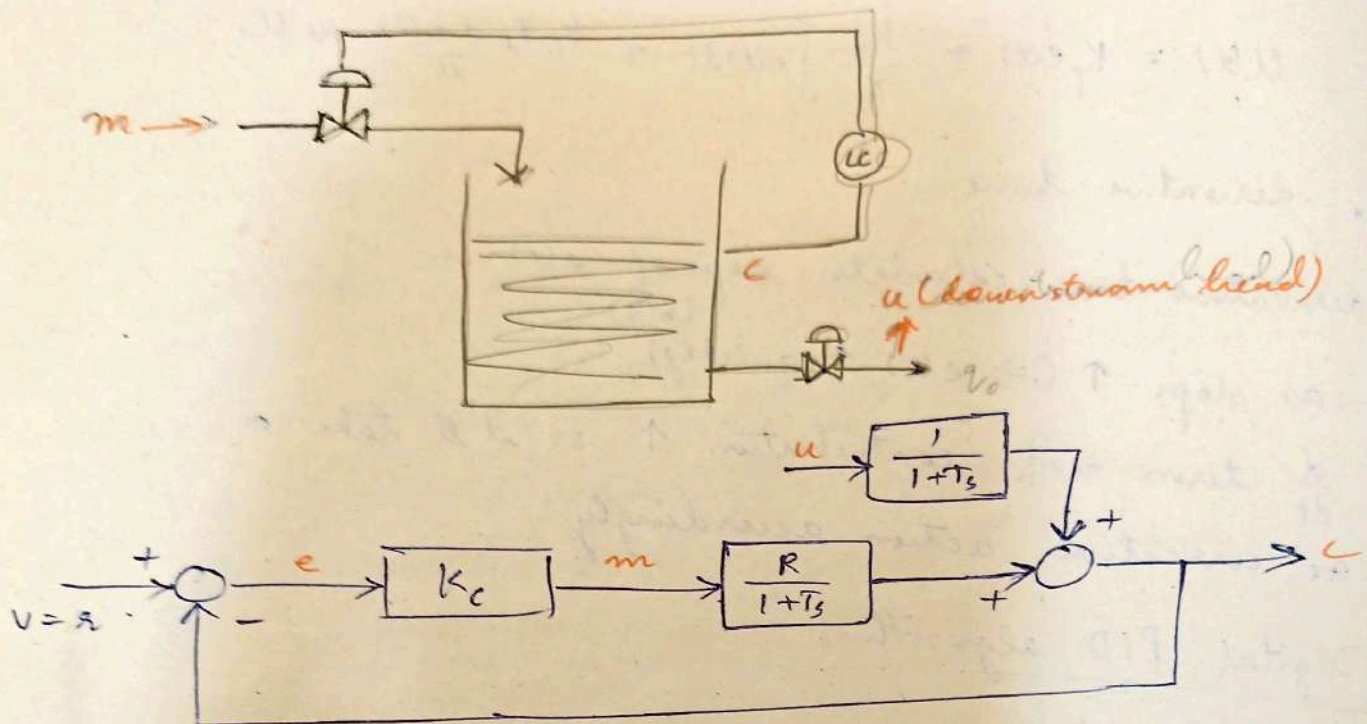
$\hookrightarrow$ abs. value of posⁿ is calculated

* velocity algo has is free from reset windup.

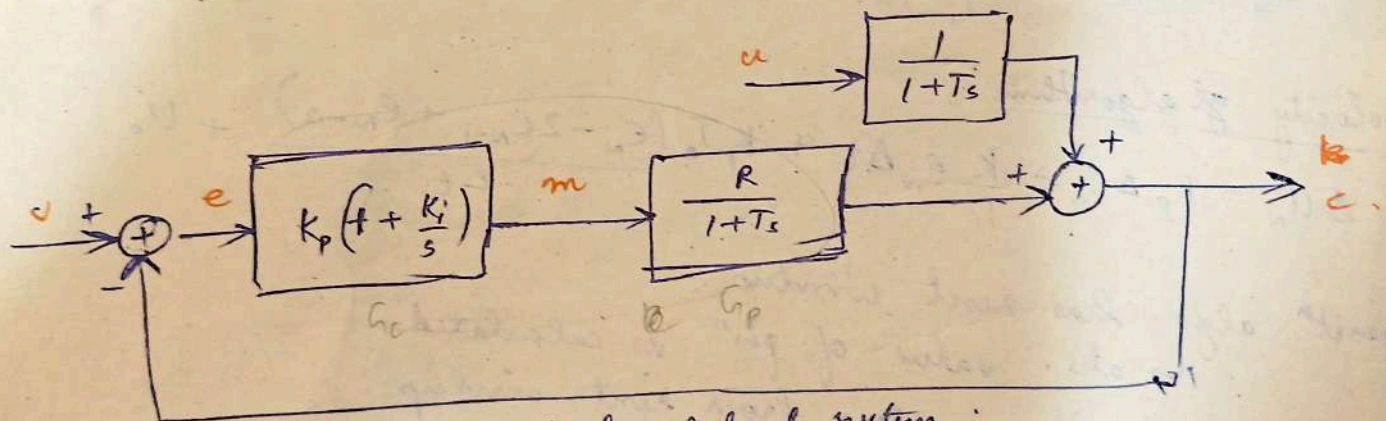
$\hookrightarrow$ change from final posⁿ is calculated.

* If else condition/situation has to be introduced to eliminate reset windup.

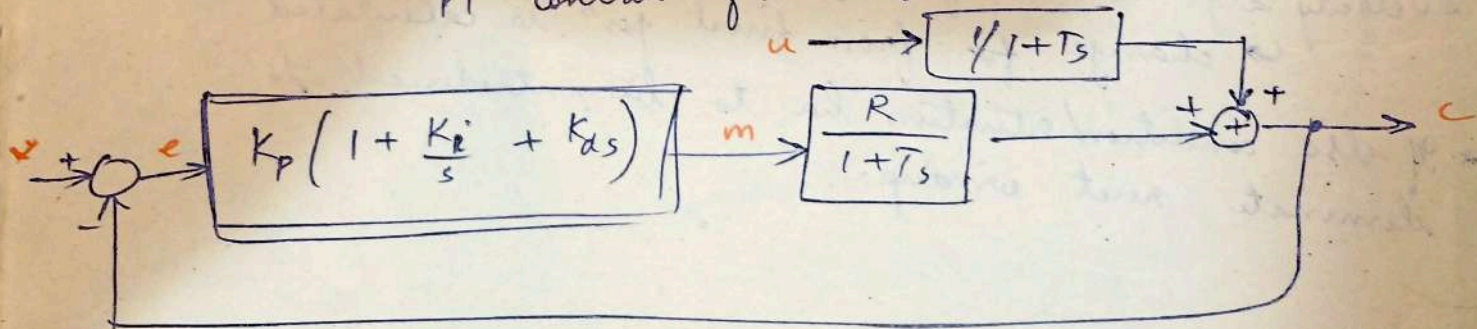
For level control system



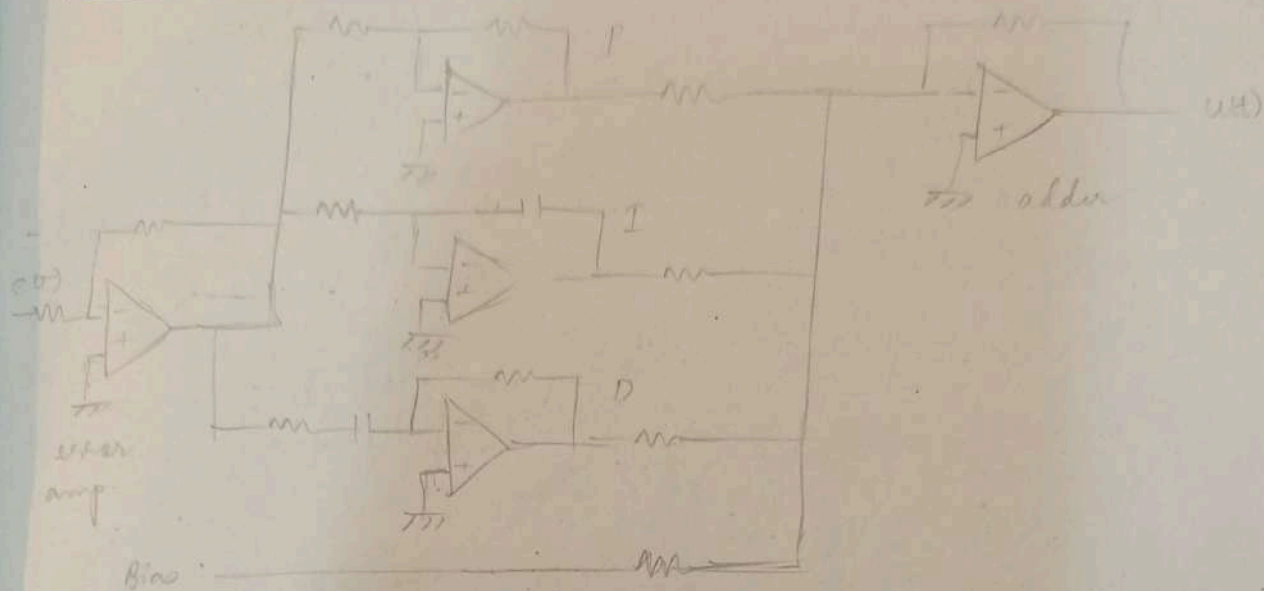
P control of level system



PI control of level system



PID control of level system



$$m = e G_c$$

$$e = v - c$$

$$m = (v - c) G_c$$

$$c = G_p m + G_d u$$

$$c = G_c G_p (v - c) + G_d u$$

let i/p to sys. be x , (not v)
o/p be y (not c)
disturbance be d (not u)
error be e

$$\Rightarrow y = G_p G_c (x - y) + G_d d$$

since want relation w/ load change,
assume $x = 0$

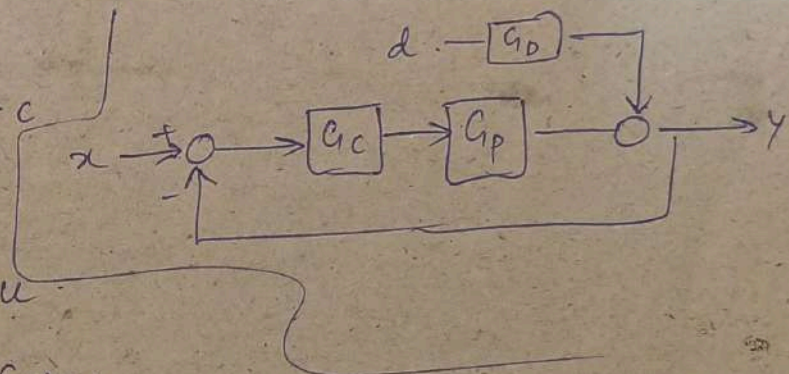
$$\therefore Y_{ss} = G_p G_c (-Y_{ss}) + G_d D_{ss}$$

$$\Rightarrow Y_{ss} (1 + G_p G_c) = G_d D_{ss}$$

$$\Rightarrow \frac{Y_{ss}}{D_{ss}} = TF = \frac{G_d}{1 + G_p G_c}$$

when load change: step,

$$Y_{ss} = \frac{G_d}{1 + G_p G_c} \cdot \frac{1}{s}$$



CL sys w/ PI controller
for load change,

$$G_c = \frac{1}{1 + T_I s}$$

$$G_p = K_p \left(1 + \frac{T_I}{s} \right)$$

$$\left(\text{PID: } K_p \left(1 + \frac{T_I}{s} + T_D s \right) \right)$$